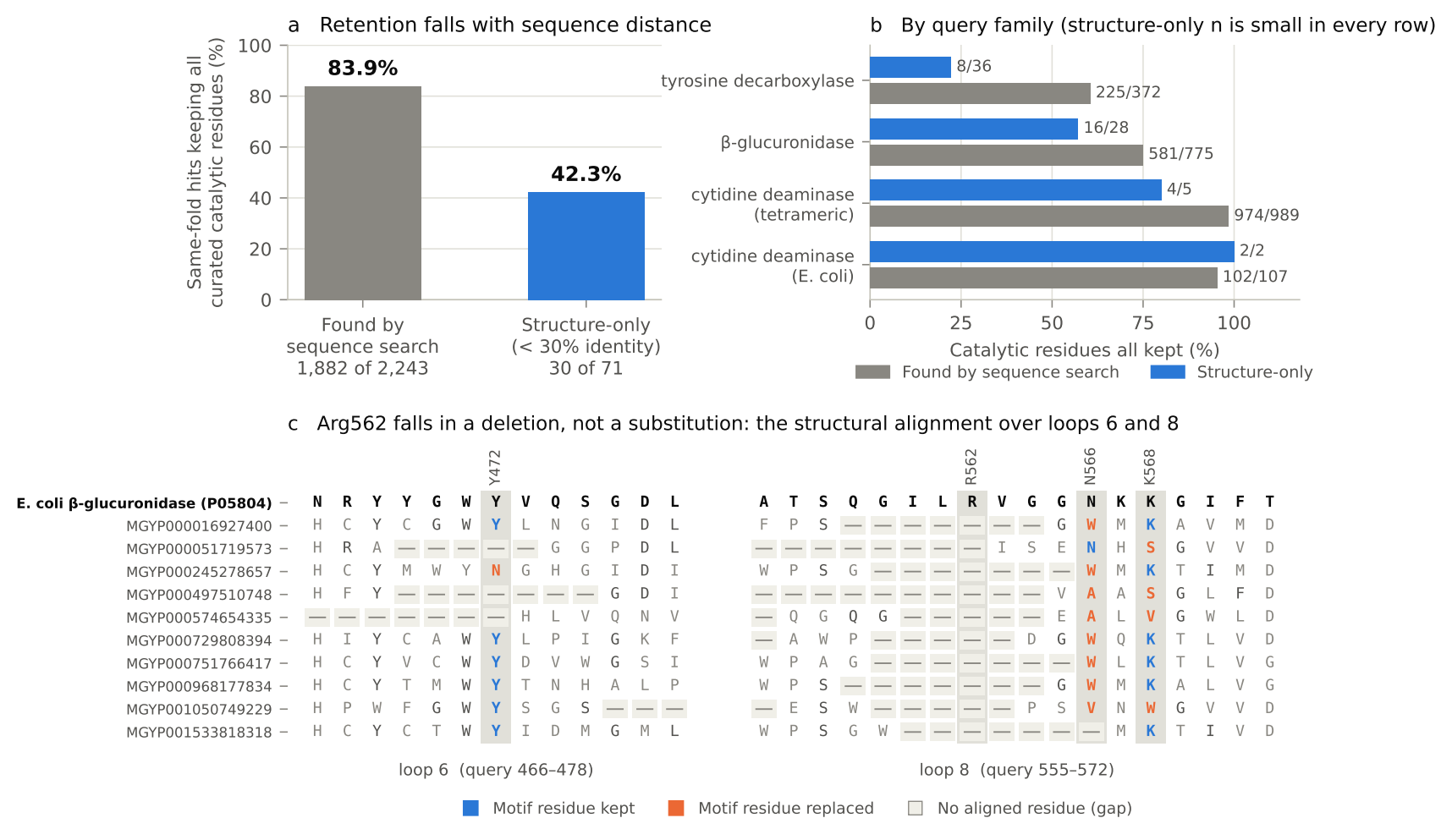
a Retention falls with sequence distance
Same-fold hits keeping all curated catalytic residues (%)
100
80
60
40
20
0
83.9%
42.3%
Found by sequence search 1,882 of 2,243
Structure-only (< 30% identity) 30 of 71
b By query family (structure-only n is small in every row)
tyrosine decarboxylase
β-glucuronidase
cytidine deaminase (tetrameric)
cytidine deaminase (E. coli)
8/36
225/372
16/28
581/775
4/5
974/989
2/2
102/107
0
25
50
75
100
Catalytic residues all kept (%)
Found by sequence search
Structure-only
c Arg562 falls in a deletion, not a substitution: the structural alignment over loops 6 and 8
Y472
R562
N566
K568
E. coli β-glucuronidase (P05804) N R Y Y G W Y V Q S G D L
MGYP000016927400 H C Y C G W Y L N G I D L
MGYP000051719573 H R A — — — — — G G P D L
MGYP000245278657 H C Y M W Y N G H G I D I
MGYP000497510748 H F Y — — — — — — — G D I
MGYP000574654335 — — — — — — — H L V Q N V
MGYP000729808394 H I Y C A W Y L P I G K F
MGYP000751766417 H C Y V C W Y D V W G S I
MGYP000968177834 H C Y T M W Y T N H A L P
MGYP001050749229 H P W F G W Y S G S — — —
MGYP001533818318 H C Y C T W Y I D M G M L
A T S Q G I L R V G G N K K G I F T
F P S — — — — — — — G W M K A V M D
— — — — — — — — I S E N H S G V V D
W P S G — — — — — — — W M K T I M D
— — — — — — — — — — V A A S G L F D
— Q G Q G — — — — — — E A L V G W L D
— A W P — — — — — D G W Q K T L V D
W P A G — — — — — — — W L K T L V G
W P S — — — — — — G W M K A L V G
— E S W — — — — — P S V N W G V V D
W P S G W — — — — — — — M K T I V D
loop 6 (query 466–478)
loop 8 (query 555–572)
Motif residue kept
Motif residue replaced
No aligned residue (gap)